Virtual Hosting in user directory

1. Deployment Objective

Host a simple PHP website in a user home directory with the following specific requirements:

1. Create a new user named Rejin
2. Create a sample web page under a directory named public_html (/home/Rejin/public_html)
3. Access the website using URL: myfirstwebsite.test and www.myfirstwebsite.test

Webpage content:

<?php
phpinfo();
?>

2. Package Installation & Service Activation

First, update your local DNF package repository cache registries before pulling new network binaries:

```
dnf update -y
```

2.1 Apache HTTP Server Deployment

Install the Apache Web Server package independently, then start and register the background tracking daemon service wrapper manually:

```
dnf install httpd -y
systemctl enable --now httpd
```

2.2 PHP Processing Engine Deployment

Install the PHP processing components alongside the corresponding FastCGI Process Manager handler independently, then register the service tracking hooks:

```
dnf install php php-fpm -y
systemctl enable --now php-fpm
```

3. User Provisions & Target Document Root Creation

Provision the targeted standalone system user identity 'Alice' and build the standard public-facing structural directory hierarchy within their environment:

```
adduser Rejin
mkdir -p /home/Rejin/public_html
```

Generate the requested static script interface containing core runtime telemetry inside the document root folder:

```
nano /home/Rejin/public_html/index.php

# Populate with the following contents:
<?php
phpinfo();
?>
```

Reassign the execution context boundaries completely over to the new user namespace recursively:

```
chown -R Rejin:Rejin /home/Rejin/public_html
```

4. SELinux Security Paradigms & Discretionary Access Controls

RHEL enforces mandatory Access Controls via SELinux. In order for the out-of-process HTTP daemon to successfully route requests and fetch assets from a non-standard staging path such as a home folder, explicit filesystem policy labels and global Booleans must be updated permanently.

Configure POSIX directory bits to allow external standard process traversal:

```
chmod 711 /home/Rejin
chmod 755 /home/Rejin/public_html
chmod 644 /home/Rejin/public_html/index.php
```

Append file-context definitions to the running SELinux sub-system mapping table and force validation across the tree context:

```
semanage fcontext -a -t httpd_user_content_t "/home/Rejin/public_html(/.*)?"
restorecon -R -v /home/Rejin/public_html
```

Toggle the persistent system boolean to explicitly permit home directory evaluation natively across Apache instances:

```
setsebool -P httpd_enable_homedirs 1
```

5. Apache Virtual Host Topography Configuration

Create a localized Virtual Host context allocation configuration module within the active configuration layout directory hierarchy:

```
vim /etc/httpd/conf.d/myfirstwebsite.conf
```

```
<VirtualHost *:80>
    ServerName myfirstwebsite.test
    ServerAlias www.myfirstwebsite.test
    DocumentRoot /home/Rejin/public_html

    <Directory /home/Rejin/public_html>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog /var/log/httpd/myfirstwebsite_error.log
    CustomLog /var/log/httpd/myfirstwebsite_access.log combined
</VirtualHost>
```

Validate the operational parameters structural state, then re-initialize the processing engine core gracefully:

```
systemctl restart httpd
```

6. Firewalls, Networking, and Local Name Resolution

To facilitate secure operational transport across environments, modifications must occur inside both the AWS Control Console and your local user machine.

6.1 AWS Inbound Security Constraints

1. Navigate to the AWS EC2 Management Console and select your running instance container.
2. Under the 'Security' subsection menu, click the active Security Group link.
3. Select 'Edit inbound rules' and specify a new tracking parameter:
   - Type: HTTP
   - Port Range: 80
   - Source Type: Anywhere-IPv4 (0.0.0.0/0)

6.2 Localized Host Manipulation File Routing

Because the domain addresses requested (myfirstwebsite.test and www.myfirstwebsite.test) do not sit inside a registered global public top-level domain namespace, name mapping must occur on your workstation local operating system hosts lookup loop table.

Locate your corresponding tracking file based on your workstation variant:

• Windows Desktop Platforms: C:\Windows\System32\drivers\etc\hosts

Append the downstream entry line mapped directly onto the destination address node instance (Replace the illustrative sample target with your live EC2 Public Address):

```
13.207.60.72    myfirstwebsite.test
13.207.60.72    www.myfirstwebsite.test
```

6.3 Output Pictures

PHP Version 8.3.29

System	Linux ip-172-31-32-235.ap-south-1.compute.internal 6.12.0-124.52.1.el10_1.x86_64 #1 SMP PREEMPT_DYNAMIC Sat Apr 11 20:17:08 EDT 2026 x86_64
Build Date	Dec 16 2025 14:32:42
Build System	Red Hat Enterprise Linux release 10.1 (Coughlan)
Build Provider	Red Hat, Inc.
Compiler	gcc (GCC) 14.3.1 20250617 (Red Hat 14.3.1-2)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmlreader.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831,NTS
PHP Extension Build	API20230831,NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	provided by mbstring
Zend Max Execution Timers	disabled
IPv6 Support	enabled
DTrace Support	available, disabled